

SOD-523 Surface Mount Schottky Barrier Rectifier

Features

- $V_R=100V$
- $I_{F(AV)}=0.15A$
- Power Dissipation of 150mW
- Fast switching speed
- Meets MSL level 1, per J-STD-020, LF maximum peak of 260°C

Applications

For use in low voltage high frequency circuit signals.

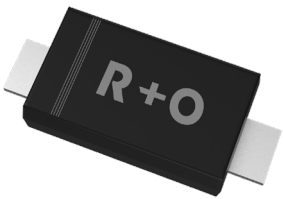
Mechanical Data

- Case: SOD-523
Molding compound meets UL 94V-0 flammability rating, RoHS-compliant, halogen-free
- Terminals: Solder plated, solderable per MIL-STD-750, Method 2026
- Polarity: Cathode line denotes the cathode end

Function Diagram

Reverse Voltage
100 V
Forward Current
0.15 Ampere

SOD-523



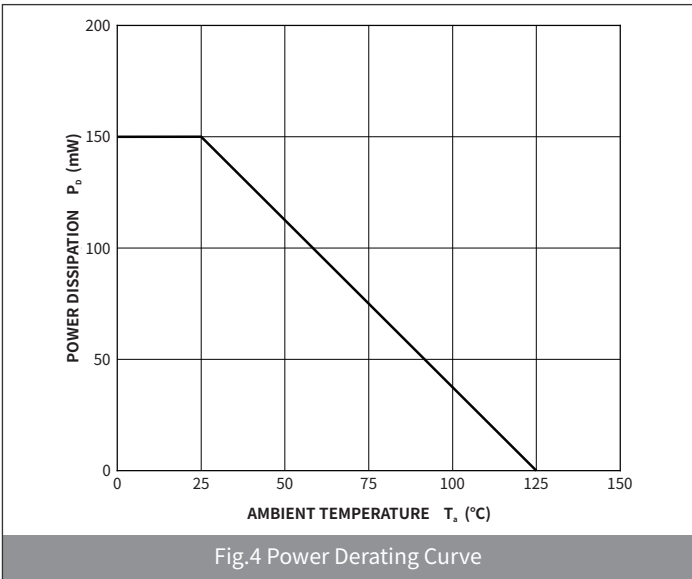
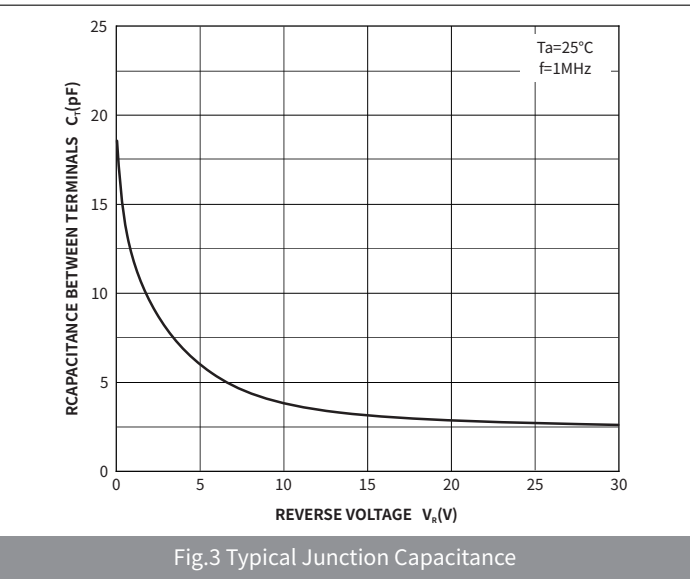
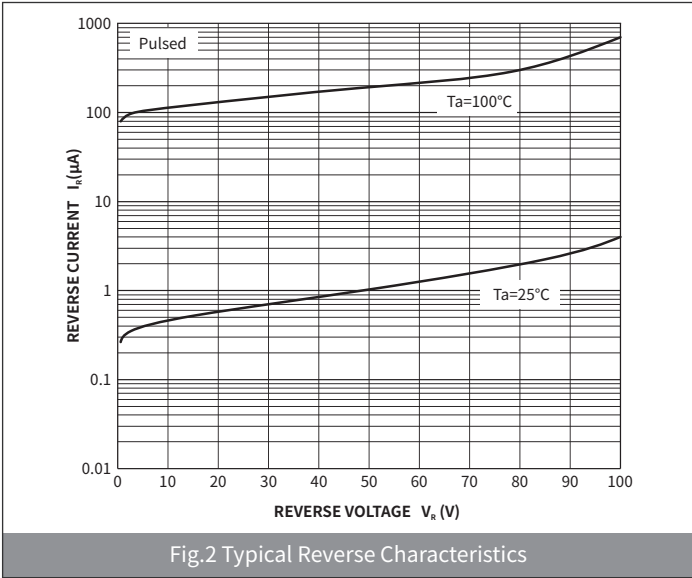
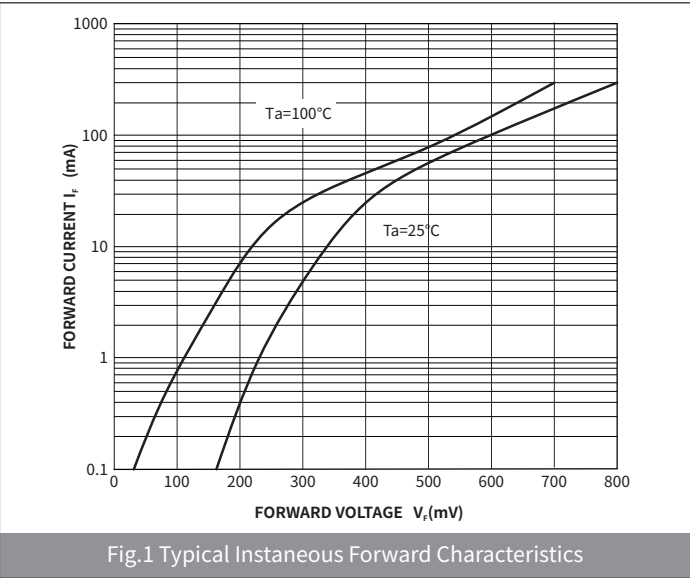
Maximum Ratings (Ta=25°C Unless otherwise specified)

PARAMETER	SYMBOL	UNIT	VALUE
Maximum repetitive peak reverse voltage	V_{RRM}	V	100
Maximum RMS Voltage	V_{RMS}	V	70
Maximum DC blocking Voltage	V_{DC}	V	100
Maximum average forward rectified current	$I_{F(AV)}$	A	0.15
Non-repetitive Peak Forward Surge Current @ t=8.3ms Half-sine wave	I_{FSM}	A	0.75
Power Dissipation	P_D	mW	150
Junction Temperature	T_j	°C	-40 ~+125
Storage temperature range	T_{STG}	°C	-55 ~+150
Typical thermal resistance	$R_{\theta JA}$	°C /W	667

● **Electrical Characteristics** (Ta=25°C Unless otherwise noted)

PARAMETER	TEST CONDITIONS	SYMBOL	UNIT	Min	Type	Max
Maximum forward voltage	$I_F=0.1\text{mA}$	V_F	V	—	—	0.25
	$I_F=10\text{mA}$			—	—	0.45
	$I_F=250\text{mA}$			—	—	1.0
Maximum reverse current	$V_R=1.5\text{V}$	I_R	μA	—	—	0.3
	$V_R=75\text{V}$			—	—	2.0
Reverse voltage	$I_R=100\mu\text{A}$	$V_{(BR)}$	V	100	—	—
Capacitance between terminals	$V_R=0\text{V}, f=1.0\text{MHz}$	C_T	pF	—	20	—
	$V_R=1\text{V}, f=1.0\text{MHz}$			—	12	—

● **Ratings And Characteristics Curves** (Ta=25°C Unless otherwise specified)



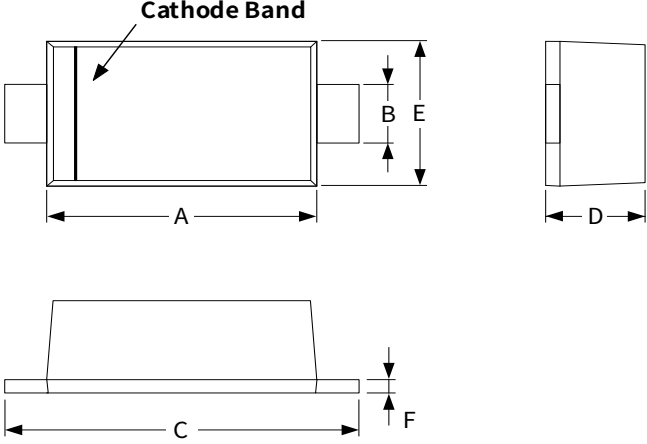
BAT46X

SURFACE MOUNT SCHOTTKY BARRIER RECTIFIER

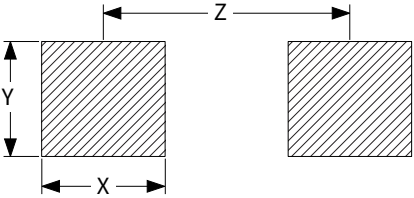
● Ordering Information

PACKAGE	PACKAGE CODE	UNIT WEIGHT(g)	REEL(pcs)	BOX(pcs)	CARTON(pcs)	DELIVERY MODE
SOD-523	R1	0.002	3000	45000	180000	7"

● Package Outline Dimensions (SOD-523)

	Symbol	Dimensions			
		Millimeters		Inches	
		Min	Max	Min	Max
A		1.10	1.30	0.043	0.051
B		0.500	0.700	0.020	0.028
C		0.250	0.350	0.010	0.014
D		0.080	0.150	0.003	0.006
E		0.750	0.850	0.030	0.033
F		1.100	1.300	0.043	0.051

● Suggested Pad Layout

	Symbol	Dimensions			
		Millimeters		Inches	
		Min.	Max.	Min.	Max.
X		0.55	0.65	0.022	0.026
Y		1.05	1.15	0.041	0.045
Z		0.75	0.85	0.030	0.033